

Discrete Mathematics
MATH1031 - Durham University

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Part I

Module Content

13 Introduction to Discrete Mathematics

13.1 Arrangements and Permutations

13.1.1 Arrangements

Definition 13.1.1.1: An arrangement is a sequence of items in which the order of the items in the sequence matters.

Definition 13.1.1.2: The length of an arrangement is the number of items in the sequence.

Standard Problem 1: How many arrangements of length k can be made from elements of a set S of size $|S| = n$?

Solution: n^k as arrangement has k places, each of which can be occupied by any one of n entries.

Example 13.1.1.1: How many four-letter “words” can be made from the standard A-Z alphabet?

Solution: $26^4 = 456,796$ words.

Standard Problem 2: How many arrangements of length k with no repeats can be made from elements of a set S of size $|S| = n$?

Solution:

$$n(n-1)(n-2)\dots(n-k+1) = \frac{n!}{(n-k)!}, \quad k \leq n \quad (1)$$

As we look at the available choices for each successive item in the list, the number of options is reduced by one each time.

Definition 13.1.1.3: The multiplication principle states that, for an arrangement of length k , if for $i \in \{1, 2, \dots, k\}$ there are n_i choices for the n^{th} item, regardless of previous choices, then the number of possible arrangements is $n_1 n_2 \dots n_k$.

13.1.2 Permutations

Definition 13.1.2.1: A permutation on a finite set S is an arrangement of length $|S|$ that uses each item from S exactly once.

Proposition 13.1.2.1: If $|S| = n$, then there are $n!$ permutations on S .

Proof: This is Standard Problem 2 with $n = k$.

□

Definition 13.1.2.2: We generalise this concept as, for a non-negative integer r , an r -permutation on S is an arrangement of r distinct elements from S . We write $P(n, r)$ for the number of r -permutations on a set of size n , which, by Standard Problem 2, we have,

$$P(n, r) = n(n-1)\dots(n-r+1) = \frac{n!}{(n-r)!}, \quad 0 \leq r \leq n \quad (2)$$

13.1.3 Counting Examples

Definition 13.1.3.1: The addition principle states that, for two disjoint, finite sets A and B . Then,

$$A \cap B = \emptyset \quad |A \cup B| = |A| + |B| \quad (3)$$

More generally, if $A_1, A_2, \dots, A_n$ are pairwise disjoint, then

$$|A_1 \cup A_2 \cup \dots \cup A_n| = |A_1| + |A_2| + \dots + |A_n| \quad (4)$$

13.2 Combinations

Definition 13.2.0.1: A combination from a set of objects is an unordered selection of objects of that set.

Standard Problem 3: How many combinations without repetition do we have of a given size k from a set of size n ?

Solution: First consider that there are $P(n, k) = \frac{n!}{(n-k)!}$ arrangements without repetition. These are ordered lists, sorting them into groups of lists using the same letters will give groups of size $k!$. This means that the number of combinations is

$$\frac{P(n, k)}{k!} = \frac{n!}{k!(n-k)!} \quad (5)$$

Using the binomial coefficient notation, this means the number of combinations is $\binom{n}{k}$.

Example 13.2.0.1: How many ways are there of selecting ten different letters of the alphabet?

Solution: $\binom{26}{10} = \frac{26!}{10!16!}$.

Proposition 13.2.0.1: $\binom{n}{k} = \binom{n}{n-k}$

Proof: This can be seen from the formula for the binomial. □

Proposition 13.2.0.2: Pascal's formula, for $1 \leq k \leq (n-1)$, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

Proof: Proof excluded. □

Theorem 13.2.0.1: The binomial theorem, if n is a positive integer and $a, b \in \mathbb{R}$, then

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \quad (6)$$

Proof: The expression $(a+b)(a+b)\dots(a+b)$ when multiplied out, gives various terms of the form $a^k b^{n-k}$. For fixed k , the number of terms $a^k b^{n-k}$ is $\binom{n}{k}$ since one chooses the k brackets in the product form from which to choose the factor a rather than b . □

13.3 Arrangements and Combinations with Repetitions

13.3.1 Arrangements with Repetitions

Standard Problem 4: How many arrangements are there of a list of objects in which some objects are repeated?

Solution: **Theorem 13.3.1.1:** Given a list of n objects of r different types, in which objects of type i appear n_i times, the number of arrangements is

$$P(n; n_1, n_2, \dots, n_r) = \frac{n!}{n_1! n_2! \dots n_r!} \quad (7)$$

Proof: Proof omitted. □

Example 13.3.1.1: How many arrangements of REARRANGE are there, and of those, how many of the arrangements have at least two R's together?

Solution: The letters to arrange are RRR AA EE G and N. Applying Theorem 13.3.1.1 with $n = 9$ and $n_1 = 3, n_2 = 2, n_3 = 2, n_4 = 1, n_5 = 1$, we get

$$\frac{9!}{3!2!2!} = 15,120 \quad (8)$$

We can then take all the cases where all the R's are separate. To do this, we arrange all the other letters, giving $\frac{6!}{2!2!}$ arrangements. We then insert the R's into the available gaps, giving 7 possible spots for 3 R's, meaning the number of arrangements are

$$\frac{9!}{3!2!2!} - \binom{7}{3} \left(\frac{6!}{2!2!} \right) = 8,820 \quad (9)$$

13.3.2 Combinations with Repetitions

Standard Problem 5: Given a set of n objects, how many combinations of size k can be chosen if objects may be selected more than once? (How many k -combinations with repetitions from n objects are there?)

Solution: To solve this problem, first consider

Standard Problem 6: How many k -combinations with repetitions from n objects are there in which each object is chosen at least once in the combination?

Solution: **Theorem 13.3.2.1:** The number of k -combinations from n different objects with repetition and each object occurring at least once is $\binom{k-1}{n-1}$ for $k \geq n$.

Proof: This question clearly has a non-zero answer only if $k \geq n$. Since order does not matter in our combinations, we represent each combination with its items in non-decreasing order. Now insert markers into the strings at the points where the numbers change. Because we have specified that each number appears at least once, there must be $(n-1)$ markers occupying $(k-1)$ possible positions in each of these lists.

□

Example 13.3.2.1: Count the 5-combinations with repetitions from $\{1, 2, 3\}$ in which each number appears at least once.

Solution: Using the method above, we get the following combinations:

$$\begin{aligned} 11123 &\leftrightarrow 111|2|3 \\ 11223 &\leftrightarrow 11|22|3 \\ 11233 &\leftrightarrow 11|2|33 \\ 12223 &\leftrightarrow 1|222|3 \\ 12233 &\leftrightarrow 1|22|33 \\ 12333 &\leftrightarrow 1|2|333 \end{aligned}$$

There are 4 possible gaps for 2 markers, so the number of ways of arranging the markers is $\binom{4}{2} = 6$.

We can adapt this solution to solve Standard Problem 5 by allowing the markers to also be placed next to one another, or outside of the item strings. Taking this into account, we get:

Theorem 13.3.2.2: The number of k -combinations from n different objects with repetition is

$$\binom{k+n-1}{n-1} = \binom{k+n-1}{k} \quad (10)$$

Proof: As explained above.

□

13.3.3 The Extended Binomial Coefficients

Proposition 13.3.3.1: The principle of upper negation states that, if n and k are positive integers, then

$$\binom{-n}{k} = (-1)^k \binom{k+n-1}{k} = (-1)^k \binom{k+n-1}{n-1} \quad (11)$$

Proof:

$$\binom{-n}{k} = \frac{(-n)(-n-1)(-n-2)\dots(-n-k+1)}{k!} \quad (12)$$

Factoring out a factor of -1,

$$\binom{-n}{k} = (-1)^k \frac{(n)(n+1)(n+2)\dots(n+k-1)}{k!} \quad (13)$$

Which simplifies to give the claimed result.

□

14 Combinatorics

14.1 The Three Principles

14.1.1 Mathematical Induction

Definition 14.1.1.1: Mathematical induction provides a method for establishing the truth of a sequence of statements, each of which is associated with a non-negative integer. In its simplest form, it gives efficient proofs of identities such as the binomial formula.

Theorem 14.1.1.1: The principle of mathematical induction states that: suppose that $P(n)$ is a sequence of propositions labelled by integers $n \geq n_0$. Suppose that

- a. $P(n_0)$ is true, and
- b. For any $n \geq n_0$, the truth of $P(n)$ implies the truth of $P(n+1)$.

Then, for every $n \geq n_0$, $P(n)$ is true.

Proof: $P(n_0)$ is true by assumption (a). Then by (b), $P(n_0+1)$ is also true. Again, by (b), $P(n_0+2)$ is also true, and so on. This chain of deductions eventually reaches any given $n \geq n_0$, so $P(n)$ is true for all $n \geq n_0$.

□

14.1.2 The Pidgeon-Hole Principle

Definition 14.1.2.1: The pidgeon-hole principle states that if n envelopes are distributed among m pidgeon-holes, and $n \geq m$, then at least one pidgeon-hole contains more than one envelope.

14.1.3 The Inclusion-Exclusion Principle

Proposition 14.1.3.1: We have

$$|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3| \quad (14)$$

Proof: Apply $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$ repeatedly.

□

Theorem 14.1.3.1: The inclusion-exclusion formula states that, for a positive integer n and finite sets $A_1, \dots, A_n$,

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{k=1}^n \left((-1)^{k+1} \times \sum_{i_1 < i_2 < \dots < i_k} |A_{i_1} \cup A_{i_2} \cup \dots \cup A_{i_k}| \right) \quad (15)$$

Proof: A slightly tricky induction on n , with the base case $n = 2$ given by $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$. □

Definition 14.1.3.1: A derangement of the numbers $\{1, 2, \dots, n\}$ is a permutation in which no number is in its correct position.

Proposition 14.1.3.2: The number of derangements of $1, 2, \dots, n$ is given by

$$d(n) = \sum_{k=0}^n (-1)^k \frac{n!}{k!} \quad (16)$$

Proof: We subtract from $n!$ the number of non-derangements to get

$$d(n) = n! \sum_{k=0}^n (-1)^k \frac{1}{k!} \quad (17)$$

□

In particular, the proportion of permutations that are derangements is

$$\frac{d(n)}{n!} = \sum_{k=0}^n (-1)^k \frac{1}{k!} \quad (18)$$

This is the power series expansion of e^{-1} , so

$$\lim_{n \rightarrow \infty} \frac{d(n)}{n!} = \frac{1}{e} \quad (19)$$

14.2 Recurrence Relations

14.2.1 Basic Definitions

Definition 14.2.1.1: As defined by (Perepelitsa, 2020, p.1), a recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence for all integers n with $n \geq n_0$, where n_0 is a non-negative integer.

Definition 14.2.1.2: As defined by (Perepelitsa, 2020, p.1), a sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation.

Definition 14.2.1.3: As defined by (Perepelitsa, 2020, p.1), the initial conditions for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

Let $\{a_n\}$ be a sequence indexed by integers $n = 0, 1, 2, \dots$, we are given:

- The initial conditions $a_0, a_1, \dots, a_{r-1}$.
- The recurrence relation

$$a_n = \sum_{i=1}^r c_i a_{n-i} = c_1 a_{n-1} + \dots + c_r a_{n-r}, \quad (n \geq r) \quad (20)$$

where $c_1, \dots, c_r$ are specified constants.

Definition 14.2.1.4: The recurrence relation is linear because no term involves more than a single factor of the a_i , and it is homogeneous because every term involves an a_i .

Definition 14.2.1.5: We seek solutions to the recurrence relation (20) of the form $a_n = x^n$, $x \neq 0$. Substituting this into the relation, we get

$$x^n = \sum_{i=1}^r c_i x^{n-i}, \quad (n \geq r) \quad (21)$$

Dividing through by x^{n-r} , we get

$$x^r - \sum_{i=1}^r c_i x^{r-i} = 0 \quad (22)$$

This is the characteristic equation.

14.2.2 General Method for Solving Homogeneous Linear Recurrence Relations

1. Write down the characteristic equation of the recurrence relation as in (22).
2. Find the equation's roots $\alpha_1, \alpha_2, \dots, \alpha_r$.
3. Then, for each i , $a_n = d(\alpha_i)^n$ satisfies the recurrence relation (20) for any constant d .
4. By linearity, any combination of the $(\alpha_i)^n$ also solves the recurrence relation.
5. Find the general solution to the recurrence relation.

- If the roots are distinct, then the general solution to the recurrence relation is:

$$a_n = g(n) = d_1(\alpha_1)^n + d_2(\alpha_2)^n + \dots + d_r(\alpha_r)^n \quad (23)$$

for arbitrary constants $d_1, d_2, \dots, d_r$.

- If α is a root repeated k times, then in place of the expression (23), we use instead

$$a_n = g(n) = (e_1 + e_2 n + \dots + e_k n^{k-1})(\alpha_1)^n + d_2(\alpha_2)^n + \dots + d_r(\alpha_r)^n \quad (24)$$

for constants $e_1, e_2, \dots, e_k$.

6. Apply the initial conditions, giving r linear equations in the r unknown constants d_i . Solve these to get a solution a_n .

Example 14.2.2.1: Solve $a_n = 3a_{n-1} + 4a_{n-2}$, ($n \geq 2$) where $a_0 = 1$ and $a_1 = 0$.

Solution: The characteristic equation is

$$x^2 - 3x - 4 = 0 \quad \rightarrow \quad (x - 4)(x + 1) = 0 \quad (25)$$

This has distinct roots $x = 4$ and $x = -1$, so the general solution is

$$a_n = d_1(-1)^n + d_2(4)^n \quad (26)$$

Using the initial conditions gives

$$1 = a_0 = d_1 + d_2 \quad \text{and} \quad 0 = a_1 = -d_1 + 4d_2$$

Hence $d_1 = \frac{4}{5}$ and $d_2 = \frac{1}{5}$, so the solution is

$$a_n = \frac{1}{5}(4^n) + \frac{4}{5}(-1)^n \quad (27)$$

14.2.3 General Method for Solving Inhomogeneous Linear Recurrence Relations

Redefine the recurrence relation (20) to include an inhomogeneous part $f(n)$,

$$a_n = c_1 a_{n-1} + \dots + c_r a_{n-r} + f(n) \quad (28)$$

1. Find the general solution to the homogeneous problem $g(n) = a_n$.
2. Find a specific particular solution $p(n) = a_n$ to the full inhomogeneous problem using Table 1.
 - If $f(n)$ contains q^n where q is a root of the characteristic equation, then q^n should be replaced by nq^{n-1} .
 - If 1 is a root of the characteristic equation, then $f(n) = An^k$ should be considered as $f(n) = An^k \cdot 1^n$ and so should be treated as above.
3. The general solution to the inhomogeneous problem is then the sum of these

$$a_n = g(n) + p(n) \quad (29)$$

4. Apply the initial conditions to determine the unknown constants.

f(n)	Form of Particular Solution $p(n)$
A	B_0 (constant)
An	$B_0 + B_1 n$ (linear)
An^2	$B_0 + B_1 n + B_2 n^2$ (quadratic)
Aq^n	$B_0 q^n$ (exponential)
Anq^n	$(B_0 + B_1 n)q^n$ (etc...)

Table 1: Forms of the Particular Solution

Example 14.2.3.1: Solve $a_n = a_{n-1} + 2a_{n-2} - 2n$, ($n \geq 2$) where $a_0 = a_1 = 0$.

Solution: Solving the homogeneous recurrence relation, we get the characteristic equation

$$x^2 - x - 2 = 0 \quad \rightarrow \quad (x - 2)(x + 1) = 0 \quad (30)$$

Which has distinct roots $x = -1$ and $x = 2$, giving the general solution

$$g(n) = d_1(-1)^n + d_2(2)^n \quad (31)$$

The inhomogeneous part is $f(n) = -2n$, so we seek a particular solution of the form

$$p(n) = B_0 + B_1 n \quad (32)$$

Setting $a_n = p(n)$ in the inhomogeneous recurrence relation gives

$$B_1 n + B_0 = (B_1(n - 1) + B_0) + 2(B_1(n - 2) + B_0) - 2n \quad (33)$$

Which we solve to get $B_1 = 1$ and $B_0 = \frac{5}{2}$, meaning the particular solution is

$$p(n) = n + \frac{5}{2} \quad (34)$$

This gives us the general solution to the inhomogeneous problem to be

$$a_n = g(n) + p(n) = d_1(-1)^n + d_2(2)^n + n + \frac{5}{2} \quad (35)$$

Substituting in the initial conditions, we get that

$$a_n = -\frac{1}{2}(-1)^n - 2^n + n + \frac{5}{2} \quad (36)$$

14.3 Generating Functions

Definition 14.3.0.1: Generating functions encode combinatorial information into an analytical form so we can solve counting problems. For example, the ordinary generating function for the sequence $a_0, a_1, \dots$ is

$$f(x) = \sum_{k=0}^{\infty} a_k x^k \quad (37)$$

14.3.1 Counting with Generating Functions

Example 14.3.1.1: How many solutions are there to the equation

$$e_1 + e_2 + \dots + e_n = k \quad (38)$$

where each $e_i \in \{0, 1\}$ and n, k are given, positive integers?

Solution: When constructing our generating function, each e_i is associated with the expression $(x^0 + x^1) = (1 + x)$, the generating function is then the n -fold product,

$$f(x) = (1 + x) \cdot (1 + x) \dots (1 + x) = (1 + x)^n \quad (39)$$

If we expand the generating function as a power series, we get

$$(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k \quad (40)$$

Each term x^k corresponds to a choice of x^{e_1} from the first bracket, x^{e_2} from the second bracket, etc., in which the total power of x is $e_1 + \dots + e_n = k$, i.e., a solution to (38).

Lemma 14.3.1.1: For $n \geq 0$, $n \in \mathbb{Z}$, the generating function for the sequence $1, 1, \dots, 1, 0, 0, \dots$ (first $(n + 1)$ terms are 1) is

$$\sum_{k=0}^n x^k = \frac{1 - x^{n+1}}{1 - x} \quad (41)$$

Proof: This is the sum of a geometric series which can be proven directly in the form

$$(1 - x) \sum_{k=0}^n x^k = \sum_{k=0}^n (x^k - x^{k+1}) = 1 - x^{n+1} \quad (42)$$

since the sum telescopes.

□

Lemma 14.3.1.2: For $n \geq 0$, $n \in \mathbb{Z}$, the generating function for the sequence $1, 1, 1, \dots$ is

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1 - x} \quad (43)$$

Proof: The sum to infinity makes analytical sense only for $|x| < 1$, in which case we can take $n \rightarrow \infty$ in (41) to get (43).

□

Lemma 14.3.1.3: For $n \geq 0$, $n \in \mathbb{Z}$, the generating function for the sequence $\binom{n+k-1}{k}$, ($k \geq 0$) is

$$\sum_{k=0}^n \binom{n+k-1}{k} x^k = \frac{1}{(1-x)^n} \quad (44)$$

Proof: Recall from the extended form of the binomial theorem that

$$(1-x)^{-n} = \sum_{k=0}^{\infty} \binom{n+k-1}{k} x^k \quad (45)$$

As claimed. □

Lemma 14.3.1.4: For $n \geq 0$, $n \in \mathbb{Z}$, if $f(x) = \sum_{k=0}^{\infty} a_k x^k$ and $g(x) = \sum_{k=0}^{\infty} b_k x^k$ are the generating functions of sequences a_k and b_k respectively, then

$$f(x)g(x) = \sum_{k=0}^{\infty} c_k x^k \quad (46)$$

is the generating function of the sequence $c_k = \sum_{l=0}^k a_l b_{k-l}$.

Proof: This is just collecting terms. □

Example 14.3.1.1: Find the coefficient of x^{17} in the expression $f(x) = (1 + x + x^2 + \dots + x^{10})^5$.

Solution: We use Lemma 14.3.1.2 and Lemma 14.3.1.4 to write

$$f(x) = \left(\frac{1-x^{11}}{1-x} \right)^5 = (1-x)^{-5} (1-x^{11})^5 \quad (47)$$

Considering separately the two terms in the product, we have

$$(1-x^{11})^5 = \sum_{i=0}^5 \binom{5}{i} (-1)^i x^{11i} \quad (48)$$

And

$$(1-x)^{-5} = \sum_{j=0}^{\infty} \binom{4+j}{4} x^j \quad (49)$$

The terms in the product

$$\left(\sum_{i=0}^5 \binom{5}{i} (-1)^i x^{11i} \right) \left(\sum_{j=0}^{\infty} \binom{4+j}{4} x^j \right) \quad (50)$$

that contribute x^{17} are $(i, j) = (0, 17), (1, 6)$, so the required coefficient is

$$\binom{5}{0} \binom{21}{4} - \binom{5}{1} \binom{10}{4} = 5,985 - 1,050 = 4,935 \quad (51)$$

14.3.2 Recurrence Relations and Generating Functions

14.3.3 Partitions

14.3.4 Calculus of Generating Functions

15 Graph Theory

15.1 What is a Graph?

15.2 Eulerian Graphs

15.3 Planar Graphs

Part II

References

16 Module References

16.1 Module Professors

Introduction to Discrete Mathematics and *Combinatorics* are taught by Professor Mikhail Menshikov.

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Graph Theory is taught by Irene Ayuso-Ventura.

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16.2 Cited Works

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